

Scale-Localised Modified Gravity from the EFC Action: A Single-Parameter Forecast via $E_G(k, z)$

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Abstract. We present a single-parameter, directly falsifiable test of scale-localised modified gravity derived from the EFC relativistic action. The model modifies perturbation growth and lensing through a band-pass stiffness response $R(k)$ that peaks at a characteristic scale $k_c = 0.05 \, h/\text{Mpc}$ and vanishes in both IR and UV limits, recovering GR on super-horizon and galactic scales without additional screening. Combined with gravitational slip from the $\delta\lambda$ counter-term, this yields $\mu < 1$ (suppressed growth), $\eta > 1$ (anisotropic potentials), and $\Sigma > 1$ (enhanced lensing)—simultaneously satisfying the No-Go constraint on background modifications, BOSS full-shape $P(k)$, CMB ISW limits, and the survival window from Planck perturbation-sector tests. The model reduces to a single effective parameter ε_{tot} controlling slip amplitude, with a viable window $\varepsilon_{\text{tot}} \in [0.38, 0.42]$. The allowed parameter space is sharply constrained: deviations of ε_{tot} outside this $\sim 10\%$ window lead to incompatibility with either E_G data or CMB lensing, making the model directly falsifiable with near-term observations. We identify $E_G(k, z) = \Omega_m \Sigma / (f\mu)$ as the definitive test: EFC predicts a scale-dependent bump of $+5\text{--}7\%$ at $k \approx k_c$, testable at $2.5\text{--}3.5\sigma$ with DESI $\times$ Planck cross-correlations.

1 Introduction

The EFC perturbation sector produces three observable functions: $\mu(k, z)$ (effective Poisson coupling), $\eta(k, z)$ (gravitational slip), and $\Sigma(k, z)$ (lensing combination). The No-Go result (Figshare 31985163) constrains the background to Λ CDM, placing all EFC physics in the perturbation sector. The relativistic action (Figshare 31876324) provides the variational expressions, while the Kill-Test v6 calibrates the slip sweet spot.

This note presents the final, IR-regularised model and its observational forecast.

2 The Band-Pass Model

2.1 Stiffness Response

The stiffness response function takes a band-pass form:

$$R(k, a) = R_0 \frac{(k/k_c)^2}{(1 + (k/k_c)^2)^3} a^4, \quad (1)$$

with $R_0 = 0.510$ and $k_c = 0.05 \, h/\text{Mpc}$. This gives $R \propto k^2$ for $k \ll k_c$ (IR: GR recovered), peaks at $k \approx k_c$, and falls as k^{-4} for $k \gg k_c$ (UV: quasi-static action result). This form localises the gravitational modification to a finite band in Fourier space, ensuring recovery of GR both on super-horizon scales ($k \rightarrow 0$, where the constraint field becomes dynamical) and in the quasi-static UV limit ($k \rightarrow \infty$, where stiffness screening restores standard gravity).

2.2 Observables

$$\mu(k, z) = \frac{1}{F(1 + R(k, a))}, \quad (2)$$

$$\eta(k, z) = 1 + 2\varepsilon_{\text{tot}} \frac{(k/k_c)^2}{(1 + (k/k_c)^2)^2} a^2, \quad (3)$$

$$\Sigma(k, z) = \frac{\mu(1 + \eta)}{2}, \quad (4)$$

where $F = 1 + \alpha\bar{\phi} \approx 1.00005$ and $\varepsilon_{\text{tot}} \in [0.38, 0.42]$ controls the slip amplitude from the $\delta\lambda$ counter-term.

At the reference point ($k_c, z=0$): $\mu = 0.940$, $\eta = 1.200$, $\Sigma = 1.034$.

3 Observational Tests: Summary

Table 1: Summary of constraints at $\varepsilon_{\text{tot}} = 0.40$. The net preference is driven by improved CMB lensing fit, partially offset by mild tension in E_G . Positive $\Delta\chi^2$ indicates EFC preferred.

Test	$\Delta\chi^2$	Precision	Status
BOSS $P(k)$ shape	-0.07	$\sim 3\%/\text{bin}$	PASS
CMB ISW (low- ℓ)	—	$\sim 10\%$	PASS (8%)
CMB lensing A_{lens}	+3.3	5.5%	EFC better
KiDS shear S_{WL}	+0.04	2.5%	Neutral
E_G (current)	-1.9	$\sim 15\%$	GR preferred
Joint (net)	+1.4	—	EFC marginal

The model passes all individual tests within 2σ . The net joint $\Delta\chi^2 = +1.4$ marginally favours EFC, driven by the CMB lensing improvement.

4 The E_G Forecast

4.1 E_G Definition

The E_G statistic provides a direct, model-independent measurement of the ratio Σ/μ , isolating gravitational anisotropy without dependence on galaxy bias or σ_8 :

$$E_G(k, z) = \frac{\Omega_{m,0}}{f(z)} \frac{\Sigma(k, z)}{\mu(k, z)}. \quad (5)$$

In GR: $\Sigma = \mu = 1$, so $E_G = \Omega_m/f \approx 0.46$ (flat, scale-independent). In EFC: $\Sigma/\mu = (1 + \eta)/2 > 1$, producing a scale-dependent bump.

4.2 Prediction

Table 2: $E_G(k)$ at $z = 0.3$ for the band-pass model.

k [h/Mpc]	E_G^{GR}	E_G^{EFC}	$\Delta E_G/E_G$	Σ/μ
0.01	0.460	0.466	+1.2%	1.012
0.03	0.460	0.488	+5.9%	1.059
0.05	0.460	0.494	+7.4%	1.072
0.10	0.460	0.481	+4.6%	1.046
0.20	0.460	0.468	+1.6%	1.016

The EFC prediction is a $\sim 7\%$ bump in E_G centred at k_c . This is a unique, scale-dependent signature that Λ CDM cannot produce by rescaling σ_8 or A_s .

4.3 Forecast Precision

Current E_G measurements have $\sim 15\%$ errors per bin. DESI $\times$ Planck lensing cross-correlations are expected to achieve $\sim 5\%$ precision, yielding:

$$\text{Detection significance} \approx \frac{0.07}{0.05 \times 0.46} \approx 2.5\text{--}3.5 \sigma, \quad (6)$$

depending on survey configuration, effective redshift range, and scale binning.

This makes $E_G(k)$ the *definitive* falsification channel for the EFC band-pass model.

5 Discussion

5.1 Degeneracy Breaking

The model is partially degenerate with $\Lambda\text{CDM} + \sigma_8$ adjustment: growth alone shows only 0.2% modification. The degeneracy is broken by the growth–shear mismatch ($\sim 3\%$): growth requires $\sigma_8 \approx 0.810$ while shear requires $\sigma_8 \approx 0.850$, because $\Sigma \neq \mu$.

The scale-dependent $E_G(k)$ bump provides an independent, cleaner test. Importantly, the scale-dependent structure of $E_G(k)$ cannot be reproduced by any scale-independent rescaling of σ_8 or A_s , providing a clean separation from ΛCDM degeneracies. Tomographic evolution ($\Sigma \propto a^2$) offers a third axis, detectable with Stage IV surveys.

5.2 Physical Interpretation

The band-pass structure arises from the interplay between two EFC mechanisms: entropy stiffness (which suppresses growth via $\mu < 1$, peaking at k_c where the stiffness response R is maximal) and the $\delta\lambda$ counter-term (which generates gravitational slip $\eta > 1$, boosting the Weyl potential and hence lensing). The IR recovery ($\mu \rightarrow 1$ for $k \rightarrow 0$) reflects the transition from quasi-static to dynamical regime, where the constraint field λ can no longer respond algebraically.

5.3 Limitations

This analysis uses simplified likelihoods (diagonal covariance, single tomographic bin, approximate band powers). Full validation requires EFCLASS/MGCAMB integration with Planck TTTEEE + lensing, BOSS/DESI full-shape, and KiDS/DES tomographic cosmic shear.

6 Conclusion

The EFC band-pass model with three parameters (R_0 , k_c , ε_{tot}) represents a scale-localised modification of gravity that:

1. respects the No-Go constraint (Λ CDM background);
2. passes BOSS $P(k)$, CMB ISW, and Planck perturbation-sector tests;
3. marginally improves CMB lensing ($\Delta\chi^2 = +3.3$);
4. predicts a unique $E_G(k)$ bump at $k_c \approx 0.05 \text{ h/Mpc}$, testable at $\sim 3\sigma$ with DESI $\times$ Planck.

The model is tightly constrained but not yet excluded, with a single observable— $E_G(k)$ —providing a decisive near-term test. The entire model is determined by one effective parameter (ε_{tot}) whose viable range is $\sim 10\%$ wide; a single improved $E_G(k)$ measurement at $k \approx 0.05 \text{ h/Mpc}$ will either confirm the predicted anisotropy or falsify the model.

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